

Fanteng Meng

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1 Education

2023 – Present
Ph.D., Safety Science & Engineering
Beijing Jiaotong University (BJTU), State Key Lab of Rail Traffic Control & Safety & Transportation College (Intelligent Systems Dept.)
Advisor: Prof. Yong Qin
Enrolled via successive master-doctor program since 2023.

2022 – 2023
M.Eng., Transportation Planning & Management
Beijing Jiaotong University (BJTU)

2018 – 2022
B.Eng., Transportation
Shijiazhuang Tiedao University

2 Research Interests

UAV-based intelligent inspection; Machine vision & deep learning for rail transit; Edge computing; 3D reconstruction & point cloud analysis; Full-stack low-altitude intelligent perception systems.

3 Research Projects

P.I. for the item marked (P.I.); the rest are core participants (Core).

1. **Development and Demonstration Verification of Safety-Guarantee Technologies for Autonomous Rail Transit Systems.** (2022–2026) – National Key R&D Program (Core)
2. **Research on Unknown-Risk Identification of High-Speed Railway Surroundings from UAV Remote Sensing Imagery.** (2024–2026) – Fundamental Research Funds for the Central Universities (P.I.)
3. **In-depth Research and Application of UAV Inspection Technology for High-Speed Railway Infrastructure.** (2022–2024) – China Railway "Major Project" (Core)
4. **Construction of an Automatic UAV Intelligent Inspection System for Beijing–Shanghai HSR Infrastructure.** (2023–2024) – Beijing–Shanghai HSR "Major Project" (Core)
5. **Research on Automated Overhaul Equipment and Key Technologies for Railway Steel-Truss-Bridge Coating.** (2024–2027) – Beijing–Shanghai HSR "Major Project" (Core)
6. **Application of Autonomous-UAV Inspection to Defects and Rockfall Debris in Key Zones of Qinghai–Tibet Railway Bridges.** (2026–2027) – "Smart Sky Road" Major Project of the Qinghai–Tibet Railway (Core)

4 Selected Publications

* *Corresponding author* # *Co-first author*

1. **Meng, F.**, Qin, Y.*, Wu, Y.*, et al. "SRLF: Sparse Representation Learning Framework for Railroad Surrounding Potential Risk Perception Using UAV Imagery." *IEEE Trans. Intell. Transp. Syst.*, 2025.

2. **Meng, F.**, Qin, Y.*, Wu, Y.*, Shao, C., Yang, H., Jia, L. "Automatic Risk Level Evaluation System for Potential Environmental Hazards along High-speed Railroad Using UAV Aerial Photograph." *Expert Syst. Appl.*, 2025.
3. **Meng, F.**, Qin, Y.*, Wu, Y.*, Shao, C., Jia, L. "A Subtle Defect Recognition Method for Catenary Fastener in High-speed Railroad Using Destruction and Reconstruction Learning." *Adv. Eng. Inform.*, 2024.
4. Wu, Y.#, **Meng, F.#**, Qin, Y.*, Qian, Y.*, Xu, F., Jia, L. "UAV Imagery Based Potential Safety Hazard Evaluation for High-speed Railroad Using Real-time Instance Segmentation." *Adv. Eng. Inform.*, 2023.
5. Wu, Y.#, **Meng, F.#**, Qin, Y.*, Qian, Y.*, Liu, Z., Zhao, W. "Automated Anomaly Detection of Catenary Split Pins Using Unsupervised Learning." *Autom. Constr.*, 2024.
6. **Meng, F.**, Qin, Y.*, Cui, J., Wu, Y., Zhang, Z., Wei, S. "Unknown Risk Detection in External Environment of Railroad Using UAV Images." *Acta Aeronaut. Astronaut. Sin.*, 2025.
7. Qin, Y.*, **Meng, F.#**, Zhang, Z., et al. "Autonomous UAV-based Intelligent Inspection Framework for Rail Transit Infrastructure: A Review." *J. Beijing Jiaotong Univ.*, 2025.

5 Selected Engineering Projects

1. **Fully-Autonomous UAV Intelligent Inspection System for High-Speed Railway**

   Background & Objective: Owing to the tight maintenance-window constraints and blind spots in inspection coverage of high-speed railway infrastructure, built the domestically- and internationally-first fully-autonomous UAV intelligent inspection system for high-speed railways and scaled its application.

   Core Methodology: Designed a cloud-edge collaborative architecture: the drone-on-edge autonomously captures multimodal data of rail-transit facilities, while the cloud performs point-cloud processing and intelligent analysis over multiple asset classes, achieving integrated inspection management and control spanning command/dispatch and data backhaul.

   Outcome & Benefit: Scaled to about 100 km of line on the world's busiest Beijing–Shanghai railway; field-deployed 6 inspection UAVs and 4 automated drone nests; inspection efficiency is improved more than 10× over manual work, realizing routine automated inspection.

2. **Autonomous UAV Inspection of Qinghai–Tibet Railway Bridges & Rockfall Hazards**

   Background & Objective: Aimed at inspection difficulties of key bridge sections and rockfall hazards on hillsides around tunnel portals under extreme high altitude, low atmospheric pressure and strong wind, explored "Smart Sky Road" autonomous UAV inspection.

   Core Methodology: Fused point-cloud segmentation, intelligent flight routing and generative enhancement of defects to build an edge–cloud co-inspection recognition model for bridges and tunnels; via multi-temporal point-cloud registration, hillside feature extraction and change analysis, realized three-dimensional quantitative assessment of surrounding hillsides.

   Outcome & Benefit: Performed defect detection and rockfall/deformation inspection for key zones of long plateau bridges at an average altitude above 3000 m; formed an operationally applicable technical scheme for plateau-railway UAV inspection, supporting the long-term safe operation of the Qinghai–Tibet railway.

3. **Full-Coverage UAV Inspection & Coating-Defect Evaluation for Railway Steel Truss Bridge**

   Background & Objective: For large-span railway steel truss girder bridges where manual full-coverage of coating disease is difficult under narrow/complex under-bridge space and GNSS-denied conditions, carried out automated inspection and defect evaluation.

   Core Methodology: Designed obstacle-avoidance inspection routes covering the deck, sides

and the complex under-bridge zone (safe flight achieved through vision-based obstacle-space modelling together with obstacle-avoidance path search and execution); built an end-to-end evaluation workflow of "coating-defect recognition → quantitative grading → 3-D spatial tracing".

Outcome & Benefit: Supports long-term disease comparison and structural-safety assessment of railway steel truss bridge coating maintenance, and accumulates a reusable operation specification for UAV inspection of long-span steel bridges.

6 Patents

Granted Patents

1. Autonomous UAV Based Intelligent Inspection System for Equipment, Facilities, and the Environment along Railway Lines and Method Thereof. (US Patent)
2. Railway Environmental Hazard Identification and Risk Level Assessment Method Based on UAV Images.
3. UAV Route Generation Method for Transportation Facility Inspection under GNSS-denied Environments.
4. Cross-modal Adaptive Matching Based Online Extrinsic Calibration Method and System for LiDAR and Camera.

Published Patent Applications

1. Refined Catenary Split Pin Defect Recognition Method for Railway Catenary.
2. Coating Corrosion Detection, Assessment Method and System for Railway Steel Truss Bridges.
3. Unknown Risk Identification Method for Railway Surrounding Environment Using UAV Remote Sensing Images.

7 Honors & Awards

2024 Leading Technology Award (Outstanding Achievement), World Internet Conference — "Fully-autonomous UAV Intelligent Inspection Technology and Application for Rail Transit"

2024 Silver Award, 14th Challenge Cup National College Students' Entrepreneurship Competition (Belt & Road Intl. Invitational)

2024 Silver Award, "Qingchuang Beijing" Capital College Students Challenge Cup Entrepreneurship Competition

2022 Second Place, Beijing Jiaotong University Summer Deep Learning Championship

2021 First Prize, China Transportation Outstanding Student Award

2021 Provincial First Prize, 7th National College Students Engineering Training Integration Ability Competition

2021 Provincial Merit Student

2020 Provincial First Prize, 12th National College Students Mathematics Competition

2020 Provincial Second Prize, 7th College Students Physics Competition

8 Professional Service

Journal Reviewer *IEEE Trans. Intell. Transp. Syst.* (T-ITS), *Adv. Eng. Inform.* (AEI), *Expert Syst. Appl.* (ESWA)

9 Technical Skills

Languages: Python, C++, Matlab **Deep Learning:** PyTorch **Systems:** Linux, ROS **Vision:** OpenCV **Hardware:** DJI-PSDK